Amazon Product

Recommendation Systems

3/1/26 ~ Sydnee Sampson-Blackwell

Contents / Agenda

Business Problem and Data Overview

Context:

- Information overload makes product discovery challenging for consumers.
- Recommendation systems help users find relevant products, enhancing engagement and business growth.
- Amazon uses advanced recommendation techniques like item-to-item collaborative filtering.

Objective:

- Build a recommendation system for Amazon products based on user ratings.
- Extract meaningful insights from a large dataset of electronic product reviews.

Exploratory Data Analysis

Dataset Attributes:

- userId: Unique identifier for each user.
- productId: Unique identifier for each product.
- Rating: Rating given by a user to a product (1-5).
- timestamp: (Not used in this project).

Initial Data Filtering:

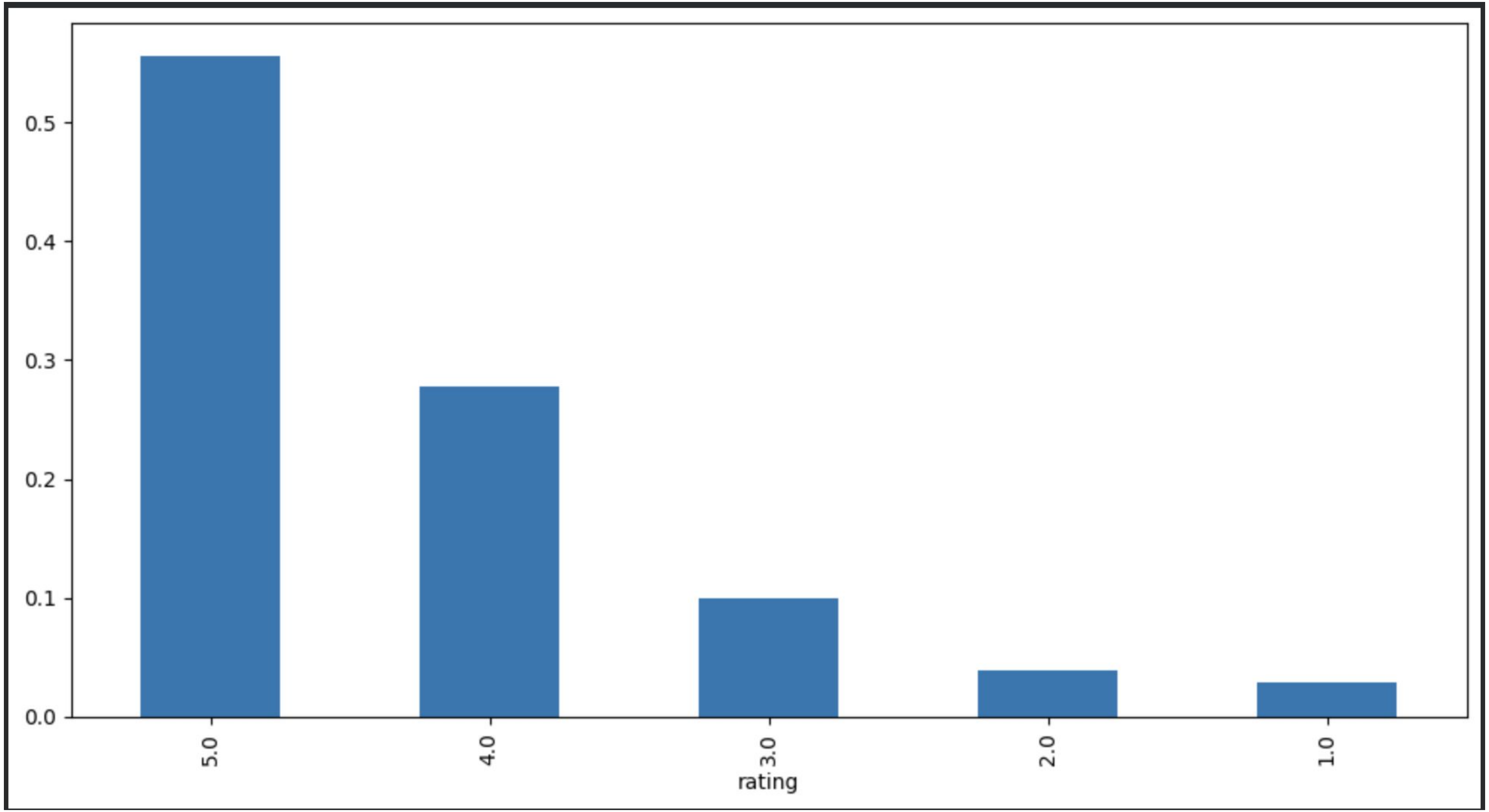
- Original data: 7,824,482 observations.
- Filtered to users with at least 50 ratings and products with at least 5 ratings.
- Final Data Shape: 65,290 rows, 3 columns.

Exploratory Data Analysis

Key EDA Observations:

- No Missing Values: All columns are complete.
- Data Types: user_id and prod_id (object), rating (float).
- Rating Distribution: Heavily skewed towards positive ratings (4s and 5s), with 5 being the most frequent rating.
- Summary Statistics (Ratings): Mean ~4.29, Median = 5.0. Indicating a strong positive bias.
- Unique Entities: 1,540 unique users, 5,689 unique products.
- Most Active User: Rated 295 products.

Rating Distribution Bar Chart



Rank Based Model

Concept:

- Recommends popular products based on overall average ratings and number of interactions.
- Lacks personalization; recommends the same products to all users.

Methodology:

1. Calculate the average rating for each product.
2. Count the number of ratings for each product.
3. Filter products based on a minimum number of interactions.
4. Sort products by average rating in descending order.

Rank Based Model

Results:

- Top 5 Products (Min 50 Interactions): B001TH7GUU, B003ES5ZUU, B0019EHU8G, B006W8U2MU, B000QUUFRW
- Top 5 Products (Min 100 Interactions): B003ES5ZUU, B000N99BBC, B002WE6D44, B007WTAJTO, B002V88HFE

Observation:

Simple and effective for general popularity. Useful for cold-start problems or as a general "best-seller" list. Does not personalize recommendations.

User-User Similarity-based Model

Concept:

- Recommends items based on preferences of similar users (people who liked similar things).
- Similarity is calculated between users.

Baseline Model Performance (Cosine Similarity):

- RMSE: 1.0012
- Precision@10: 0.855
- Recall@10: 0.858
- F1-Score@10: 0.856

User-User Similarity-based Model

Baseline Prediction Observation (User 'A3LDPF5FMB782Z', Prod '1400501466' - Actual: 5.0):

- Estimated Rating: 3.40
- Detail: was_impossible: False, actual_k: 5. (Underpredicted, but used neighbors)
- Non-Interacted Prod ('1400501466' for 'A34BZM6S9L7QI4'): Estimated Rating: 4.29. (Fallback due to was_impossible: True, reason: 'Not enough neighbors.')

User-User Similarity-based Model

Optimized Model Performance (Hyperparameter Tuned: k=40, min_k=6, Cosine Similarity):

- RMSE: 0.9526 (Improved from baseline)
- Precision@10: 0.847 (Slight decrease)
- Recall@10: 0.894 (Significant increase)
- F1-Score@10: 0.870 (Overall improvement)

Optimized Prediction Observation (User 'A3LDPF5FMB782Z', Prod '1400501466' - Actual: 5.0):

- Estimated Rating: 4.29
- Detail: was_impossible: True, reason: 'Not enough neighbors.' (Improved estimate but still a fallback)
- Non-Interacted Prod ('1400501466' for 'A34BZM6S9L7QI4'): Estimated Rating: 4.29. (Same fallback as baseline)

User-User Similarity-based Model

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Key Takeaway:

- Hyperparameter tuning improved RMSE and Recall, indicating better overall prediction accuracy and coverage.
- Still frequently resorted to fallback mechanisms for predictions due to insufficient neighbors, especially for non-interacted items, limiting true collaborative filtering application in sparse data points.

Item-Item Similarity-based Model

Concept:

- Recommends items that are similar to items a user has liked in the past.
- Similarity is calculated between items.

Baseline Model Performance (Cosine Similarity):

- RMSE: 0.9950 (Slightly better than User-User baseline)
- Precision@10: 0.838 (Slightly lower than User-User baseline)
- Recall@10: 0.845 (Slightly lower than User-User baseline)
- F1-Score@10: 0.841 (Slightly lower than User-User baseline)

Item-Item Similarity-based Model

Baseline Prediction Observation (User 'A3LDPF5FMB782Z', Prod '1400501466' - Actual: 5.0):

- Estimated Rating: 4.27
- Detail: was_impossible: False, actual_k: 22. (Closer to actual, and used neighbors)
- Non-Interacted Prod ('1400501466' for 'A34BZM6S9L7QI4'): Estimated Rating: 4.29. (Fallback due to was_impossible: True, 'Not enough neighbors.')

Item-Item Similarity-based Model

Optimized Model Performance (Hyperparameter Tuned: k=30, min_k=6, MSD Similarity):

- RMSE: 0.9576 (Improved from baseline)
- Precision@10: 0.839 (Marginal improvement)
- Recall@10: 0.880 (Significant increase)
- F1-Score@10: 0.859 (Overall improvement)

Optimized Prediction Observation (User 'A3LDPF5FMB782Z', Prod '1400501466' - Actual: 5.0):

- Estimated Rating: 4.67
- Detail: was_impossible: False, actual_k: 22. (Most accurate prediction for interacted item among KNN, used neighbors)
- Non-Interacted Prod ('1400501466' for 'A34BZM6S9L7QI4'): Estimated Rating: 4.29. (Still a fallback)

Item-Item Similarity-based Model

Key Takeaway:

- Optimized Item-Item KNN provided the most accurate prediction for an interacted item among KNN models, demonstrating effective collaborative filtering when neighbors were found.
- Similar to User-User KNN, it struggled with predictions for non-interacted items, often falling back to a default value.

Matrix Factorization based Model

Concept:

- Decomposes the user-item interaction matrix into lower-dimensional latent feature matrices for users and items.
- Captures underlying patterns in user preferences and item characteristics.
- Handles data sparsity more effectively than neighborhood-based methods.

Baseline Model Performance (random_state=1):

- RMSE: 0.8882 (Significant improvement over all KNN models)
- Precision@10: 0.853 (Competitive with best KNN)
- Recall@10: 0.880 (Strong)
- F1-Score@10: 0.866 (Highest among all baseline models)

Matrix Factorization based Model

Baseline Prediction Observation (User 'A3LDPF5FMB782Z', Prod '1400501466' - Actual: 5.0):

- Estimated Rating: 4.08
- Detail: was_impossible: False. (Made a direct collaborative prediction, no fallback)
- Non-Interacted Prod ('1400501466' for 'A34BZM6S9L7QI4'): Estimated Rating: 4.40.
- Detail: was_impossible: False. (Crucially, made a direct collaborative prediction for a non-interacted item, unlike KNN models).

Matrix Factorization based Model

Optimized Model Performance (Hyperparameter Tuned: n_epochs=20, lr_all=0.01, reg_all=0.2):

- RMSE: 0.8808 (Lowest RMSE overall, further improved)
- Precision@10: 0.854 (Excellent)
- Recall@10: 0.878 (Excellent)
- F1-Score@10: 0.866 (Maintained highest F1-Score)

Optimized Prediction Observation (User 'A3LDPF5FMB782Z', Prod '1400501466' - Actual: 5.0):

- Estimated Rating: 4.13
- Detail: was_impossible: False. (Still a direct collaborative prediction)
- Non-Interacted Prod ('1400501466' for 'A34BZM6S9L7QI4'): Estimated Rating: 4.22.
- Detail: was_impossible: False. (Consistently makes direct predictions for non-interacted items).

Matrix Factorization based Model

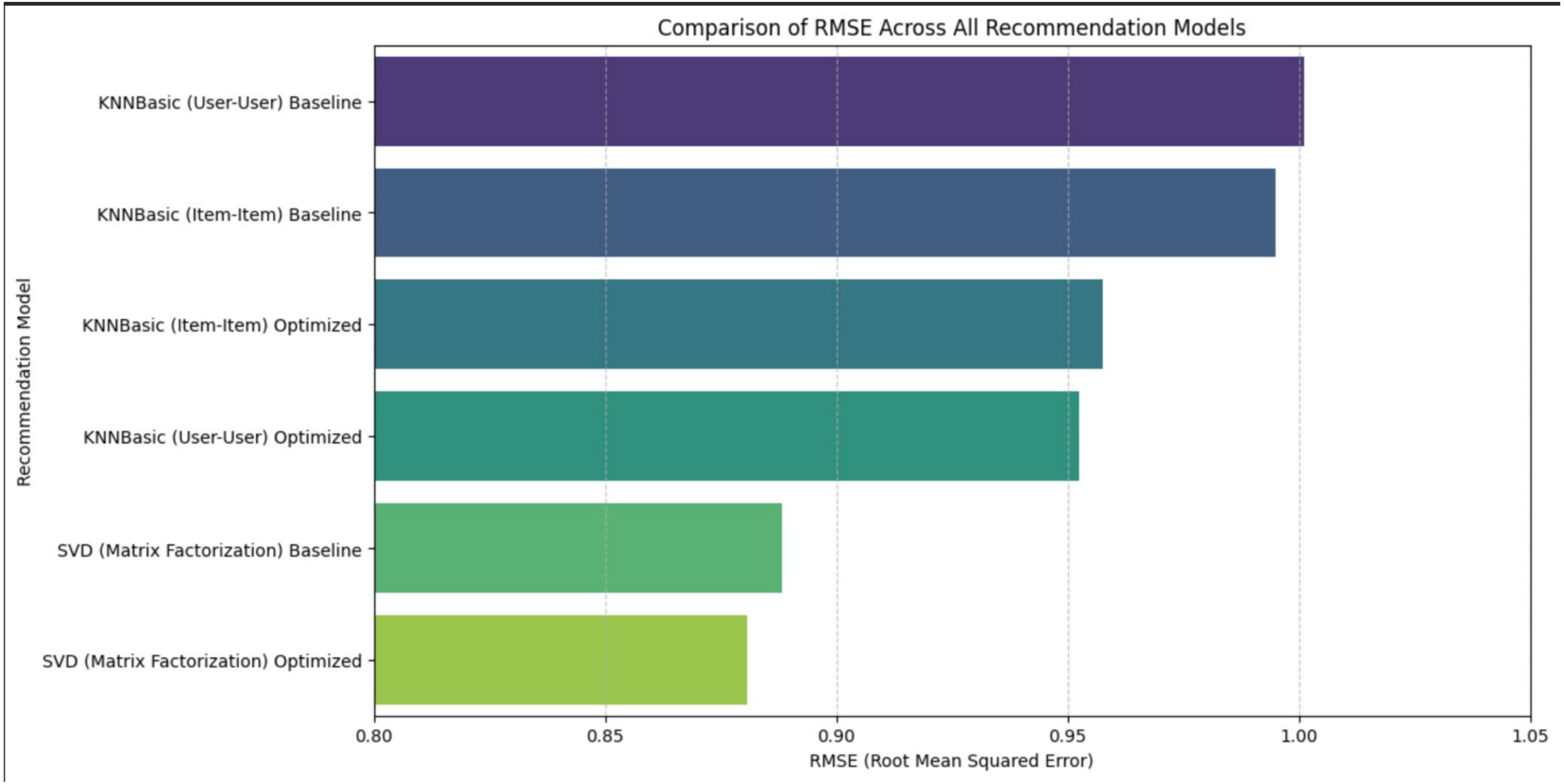
Key Takeaway:

- SVD significantly outperformed KNN models across all metrics, achieving the lowest RMSE and highest F1-score.
- Its primary advantage is the ability to make genuine collaborative predictions for both interacted and non-interacted items, effectively handling data sparsity without resorting to fallback mechanisms.

Model Comparison: Performance Summary

Model Type	Version	RMSE	Precision@10	Recall@10	F1-Score@10
Rank-Based	N/A	N/A	N/A	N/A	N/A
KNNBasic (User-User)	Baseline	1.0012	0.855	0.858	0.856
KNNBasic (User-User)	Optimized	0.9526	0.847	0.894	0.870
KNNBasic (Item-Item)	Baseline	0.9950	0.838	0.845	0.841
KNNBasic (Item-Item)	Optimized	0.9576	0.839	0.880	0.859
SVD (Matrix Factorization)	Baseline	0.8882	0.853	0.880	0.866
SVD (Matrix Factorization)	Optimized	**0.8808**	**0.854**	0.878	**0.866**

Model RMSE Comparison



Observations:

1. SVD consistently achieved the lowest RMSE and highest F1-Score, indicating the best overall prediction accuracy and balance between precision and recall.
2. Hyperparameter tuning consistently improved RMSE and F1-score for all collaborative filtering models.
3. KNN models struggled with data sparsity, often using fallback predictions for non-interacted items (was_impossible: True).
4. SVD effectively overcame data sparsity, making direct collaborative predictions for both interacted and non-interacted items (was_impossible: False).

Overall Conclusion

- The Optimized SVD (Matrix Factorization) model is the most effective for this Amazon product recommendation system.
- It provides the most accurate predictions (lowest RMSE) and a strong balance of precision and recall (highest F1-score).
- Crucially, SVD's ability to handle data sparsity and provide genuine collaborative predictions for unseen items is a significant advantage over neighborhood-based methods.

Recommendations

Implement an SVD-based core Recommendation Engine:

- Leverage the optimized SVD model for personalized recommendations due to its superior performance and robustness.

Adopt a Hybrid Approach for Cold-Start:

- For new users or products (cold-start), utilize the Rank-Based Recommendation System (popular products with high average ratings) as an initial strategy.
- Once sufficient interaction data is collected, transition to personalized SVD recommendations.

Recommendations

Continuous Improvement:

- Regular Model Retraining: Retrain the SVD model periodically with new user interaction data to maintain relevance and adapt to changing trends.
- Performance Monitoring: Continuously monitor key metrics (RMSE, Precision, Recall) to ensure the system's effectiveness.
- Explore Advanced Techniques: Investigate more sophisticated matrix factorization models (e.g., SVD++, NMF) or deep learning-based recommenders for potential further improvements.

Data Collection Strategy:

- Encourage user ratings and interactions to enrich the dataset, further enhancing the accuracy of collaborative filtering models.

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Happy Learning !